

**Problem 1a) (15 points)**

Given the image region as shown in Figure 1 and  $V = \{1\}$ , what is the shortest m-path between  $p$  (the pixel at the upper-left corner) and  $q$  (the pixel at the bottom-right corner)? Give the length of the path if the path exists.

```

1 0 1 1 0
0 1 0 1 0
0 0 1 1 0
1 1 0 1 0
0 1 1 1 1

```

**Step 1: Identify pixels in  $V$** 

$p$

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 1 | 1 |

$q$

All pixels with value 1 are highlighted. These are the only candidates for the m-path since  $V = \{1\}$ .

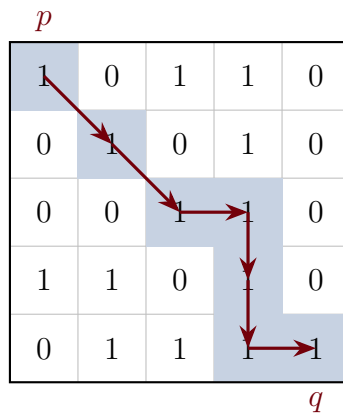
**Step 2: Apply m-adjacency rules**

Two pixels with values in  $V$  are **m-adjacent** if:

- They are 4-adjacent (share an edge), OR
- They are diagonally adjacent AND their shared 4-neighbors contain no pixels with values in  $V$ .

The second condition prevents diagonal shortcuts when a 4-connected bridge exists, eliminating path ambiguity.

### Step 3: Trace the shortest m-path



**Path:**  $(0, 0) \rightarrow (1, 1) \rightarrow (2, 2) \rightarrow (2, 3) \rightarrow (3, 3) \rightarrow (4, 3) \rightarrow (4, 4)$

**Diagonal move justifications:**

- $(0, 0) \rightarrow (1, 1)$ : Shared 4-neighbors are  $(0, 1) = 0$  and  $(1, 0) = 0$ . Neither is in  $V$ , so m-adjacent.
- $(1, 1) \rightarrow (2, 2)$ : Shared 4-neighbors are  $(1, 2) = 0$  and  $(2, 1) = 0$ . Neither is in  $V$ , so m-adjacent.

### Results

The shortest m-path exists and has **length 6**.

**Problem 1b) (15 points)**

To take a picture for fireworks at night, give at least two operations that should be applied to improve the visualization quality of the image and explain.

**Step 1: Understand the Scene Characteristics**

Fireworks at night present specific challenges:

- **High dynamic range:** Very bright fireworks against a very dark sky
- **Desired appearance:** Pure black sky with vivid, crisp firework sparks
- **No hidden information:** The dark sky contains nothing meaningful to reveal



*Original fireworks image*

## Step 2: Brightening Operations

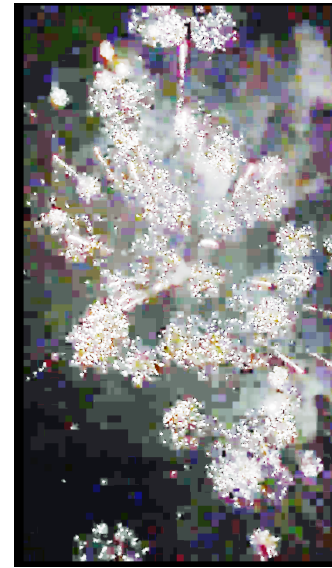
These operations attempt to reveal detail in shadows by lifting dark intensities.



Gamma ( $\gamma = 0.5$ )



Log Transform

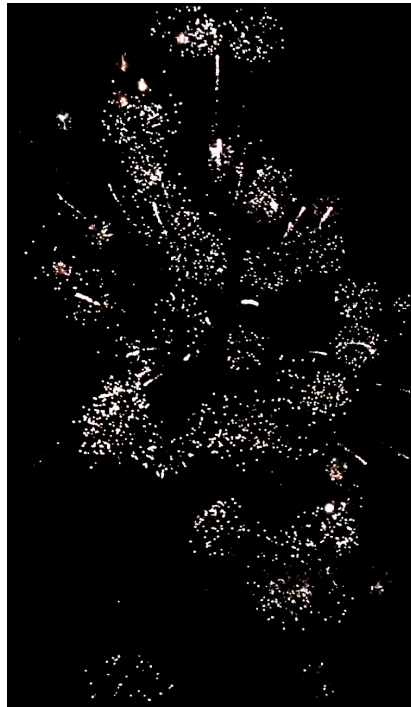


Histogram Equalization

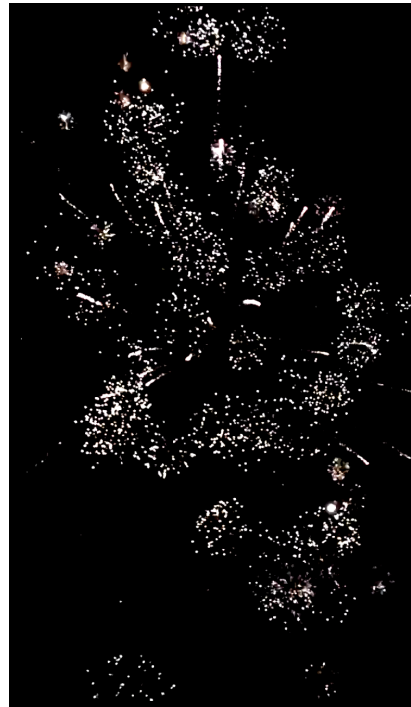
**Effect:** The black sky becomes gray, revealing noise and sensor artifacts. This is **undesirable** for fireworks because the sky *should* be black—there is no meaningful information to reveal.

### Step 3: Contrast Enhancement Operations

These operations push dark values toward black and light values toward white.



S-Curve Contrast



Boosted Contrast

**Effect:** The sky becomes pure black while the fireworks remain bright. This is **desirable** for aesthetic night photography.

### Step 4: Sharpening Operations

These operations enhance edges and fine details.



Unsharp Masking



High-Boost Filtering

**Effect:** Firework sparks appear crisper with more defined edges.

### Step 5: Color Enhancement Operations



Saturation Boost



Warm Temperature



Warm + Color Contrast

**Saturation:** Uniformly boosts all color intensity—can cause clipping in already-vivid areas.

**Warm Temperature:** Shifts colors toward orange/yellow by boosting red and reducing blue channels, enhancing the fiery appearance.

**Warm + Color Contrast:** Applies warmth first, then an S-curve in Lab color space to push warm colors warmer and cool colors cooler, creating rich golden tones.

### Step 6: Consider the Optimization Goal

The “correct” operations depend entirely on what you are trying to achieve:

| Goal                  | Recommended Operations          | Avoid                  |
|-----------------------|---------------------------------|------------------------|
| Aesthetic (fireworks) | S-curve, sharpening, saturation | Gamma, Hist EQ         |
| Surveillance          | Gamma, log transform, Hist EQ   | S-curve                |
| Medical imaging       | CLAHE, careful contrast         | Extreme transforms     |
| Object detection      | Thresholding, edge detection    | Aesthetic enhancements |

The same operation can be *beneficial* or *harmful* depending on the goal. Histogram equalization is excellent for revealing hidden details in surveillance footage, but terrible for fireworks photography where the dark background should remain dark.

### Step 7: Consider the Display Medium

The target display device also affects which operations are most beneficial:

| Display                     | Prioritize                  | Less Important        |
|-----------------------------|-----------------------------|-----------------------|
| Mobile phone (small screen) | Color, contrast, brightness | Fine sharpening       |
| Desktop monitor             | Sharpness, fine detail      | Can be subtle         |
| Print                       | Resolution, color accuracy  | Different gamma curve |

On a mobile phone, viewers cannot resolve fine detail, so aggressive sharpening provides little benefit—but punchy colors and contrast help images “pop” on a small screen viewed in variable lighting. On a desktop monitor, fine details are visible, making sharpening more valuable.

### Results

#### Recommended operations for fireworks photography:

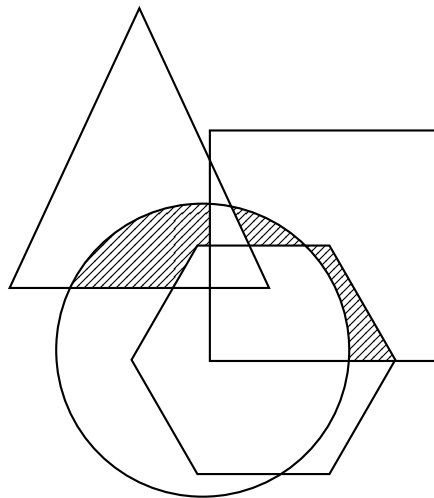
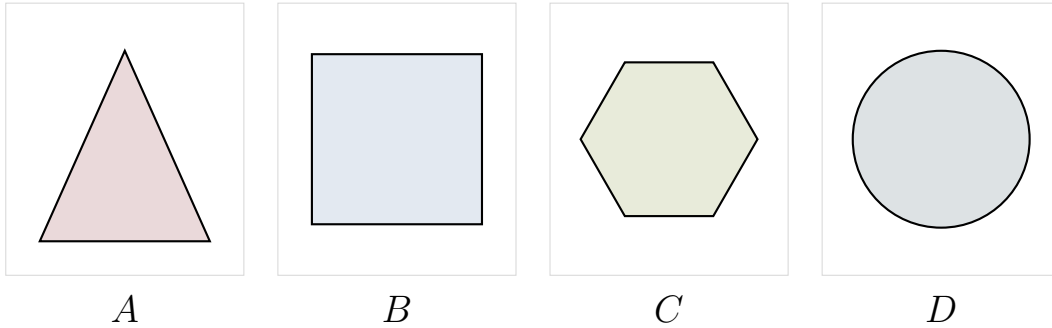
- S-curve contrast enhancement:** Apply a sigmoid transformation  $s = \frac{1}{1+e^{-k(r-0.5)}}$  to push dark pixels darker and bright pixels brighter. This ensures a pure black sky while maintaining bright firework sparks.
- Saturation boost** (especially for mobile display): Increase color saturation to make the firework colors more vivid and visually striking.
- Sharpening** (especially for desktop/print): Apply unsharp masking or high-boost filtering to enhance the crispness of firework spark edges.

**Operations to avoid:** Gamma correction ( $\gamma < 1$ ), log transforms, and histogram equalization—these lift the dark sky and reveal noise, degrading the aesthetic quality.



**Problem 2) (20 points)**

A, B, C and D are four sets. Give expressions for the set shown shaded in Figure 2 in terms of A, B, C, and D. Note that all of the shaded areas constitute of one set



**Problem 3) (25 points)**

Given a 3x3 spatial filter  $\frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$  answer the following questions:

- a) (10 points) What type of filter is it? When applying this filter on an image, what kind of effect you will expect in the output image? Why? (Hint: is this filter an image smoothing filter, an image sharpening filter or an edge detector?)

- b) (15 points) Perform the image convolution using this filter on a 3x3 image  $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$ .

Give the CROPPED filtering result for the output image. (Hint: you need to assume an appropriate zero mapping)

**Problem 4) (25 points)**

An image with intensities in the range  $r \in [0, L - 1]$  has the pdf

$$p_r(r) = \begin{cases} \frac{4r}{3(L-1)^2} & 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

where  $L - 1$  is the maximum intensity value. It is desired to transform the intensity levels of this image so that they will have the specified pdf

$$p_z(z) = \begin{cases} \frac{2z}{(L-1)^2} & 0 \leq z \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

Assume that the intensity value is continuous and find the transformation function  $z = T(r)$  in terms of  $r$  and  $z$  that will accomplish this.

(Hints: The transformation function of histogram equalization is given as

$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$

)

**Bonus Question (20 points)**

Perform the Fourier transform to a function as shown in Figure 3.

(Hint: the definition of the Fourier transform of a function is  $F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut} dt$  )

